

Software Engineering Processes

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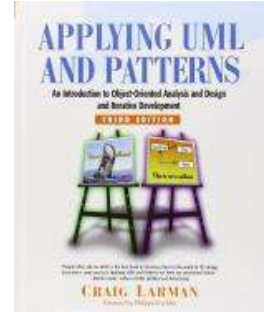
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Part 1: Unified Process

Unified Process (UP)

- ▣ An **iterative** software development process for building **object oriented** systems
- ▣ Uses OOA/D concepts
- ▣ Promotes the use of **UML** (Unified Modeling Language)
- ▣ **Very flexible** and **open**
- ▣ Uses practices from other agile methods:
 - ▣ From XP: test-driven development, refactoring, continuous integration, ...
 - ▣ From Scrum: daily meetings, ...

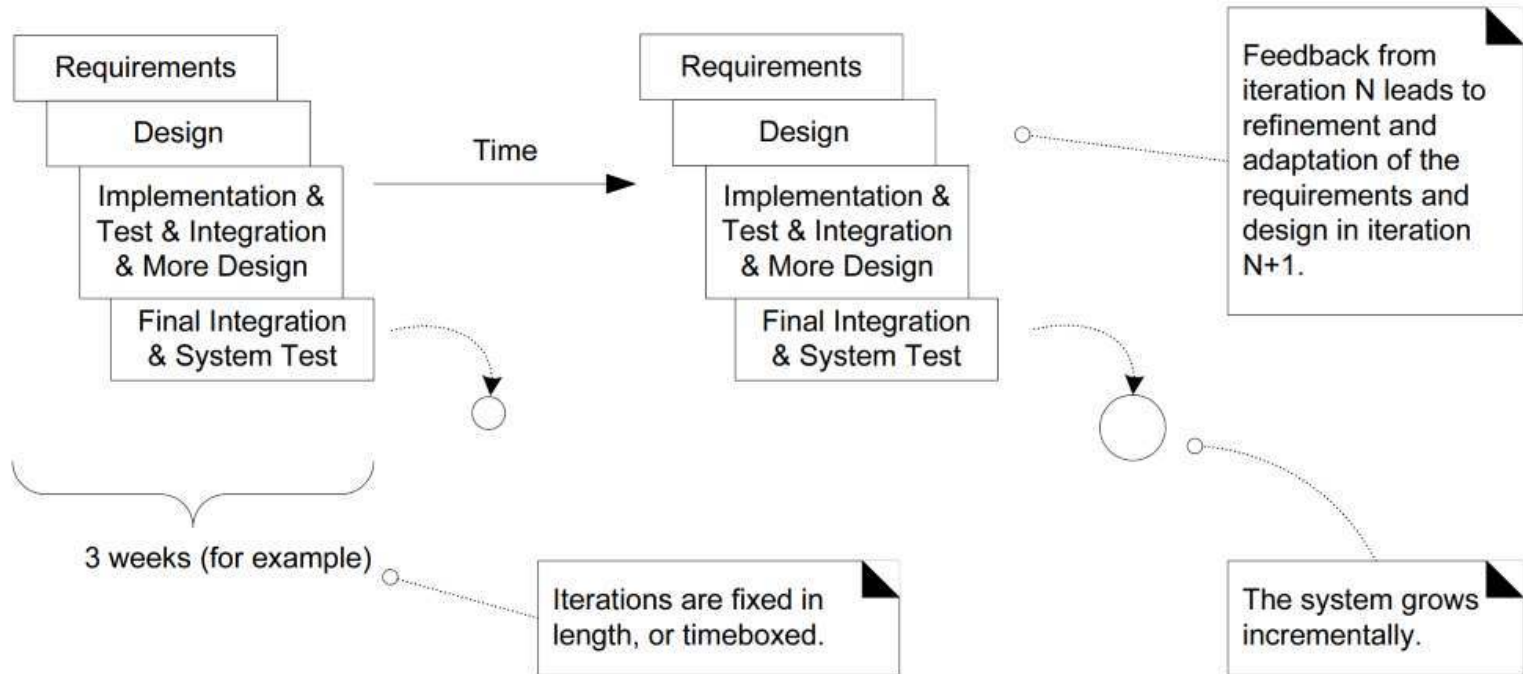
Unified Process (UP)



□ Craig Larman says:

- The **UP** combines commonly **accepted best practices**, such as an **iterative lifecycle** and **risk-driven development**, into a **cohesive** and **well-documented process** description.

Iterative, Incremental, Evolutionary Development



Iterations and Their Results

- ▣ The result of each iteration is an **executable but incomplete system**
- ▣ The result is **not ready to deliver into production**
- ▣ The system may not be eligible for production deployment until after many iterations e.g., 10 or 15 iterations or more ...
- ▣ The result of an iteration is **not an experimental or throw-away prototype**
- ▣ Iterative development is **not prototyping**
- ▣ The result is a production grade subset of the final system

Unified Process (UP) & Changes

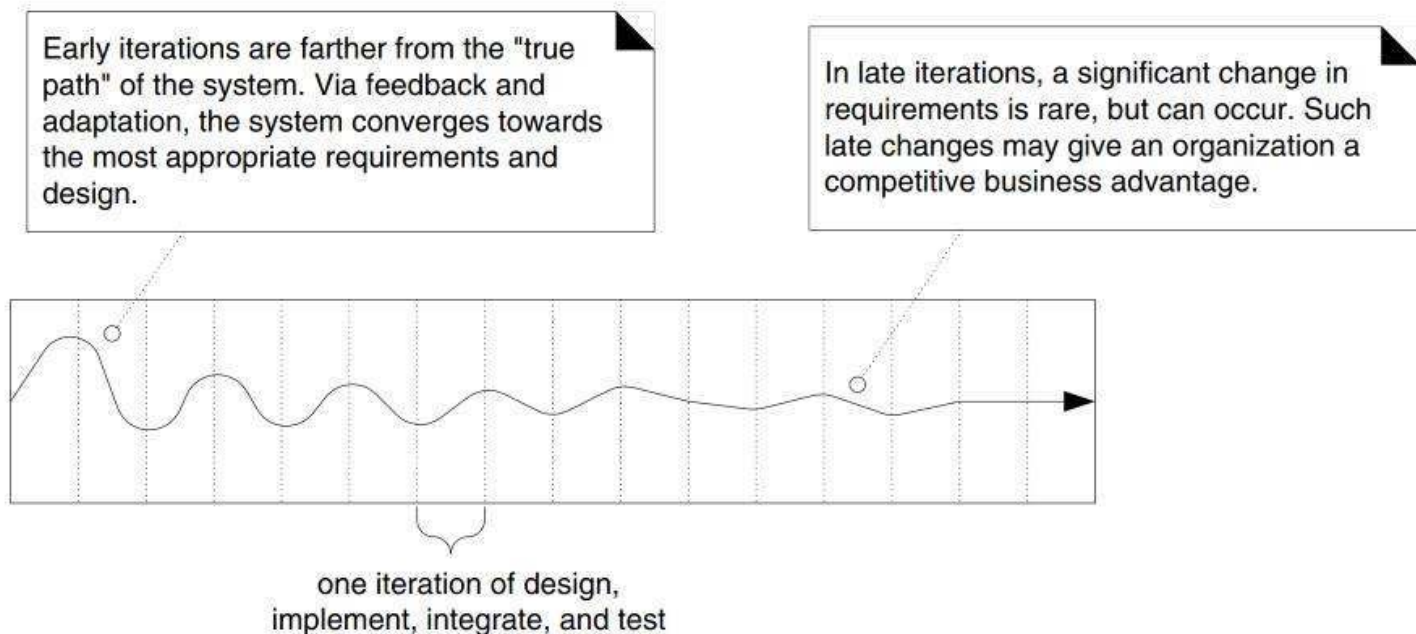
- Each iteration focusses on a **small subset of the requirements**, and **quickly designing, implementing, and testing**
- In early iterations the choice of requirements and design may not be exactly what is ultimately desired
- But taking a small step, before all requirements are finalized, or the entire design is speculatively defined, leads to rapid feedback from the users, developers, and tests

Unified Process (UP)

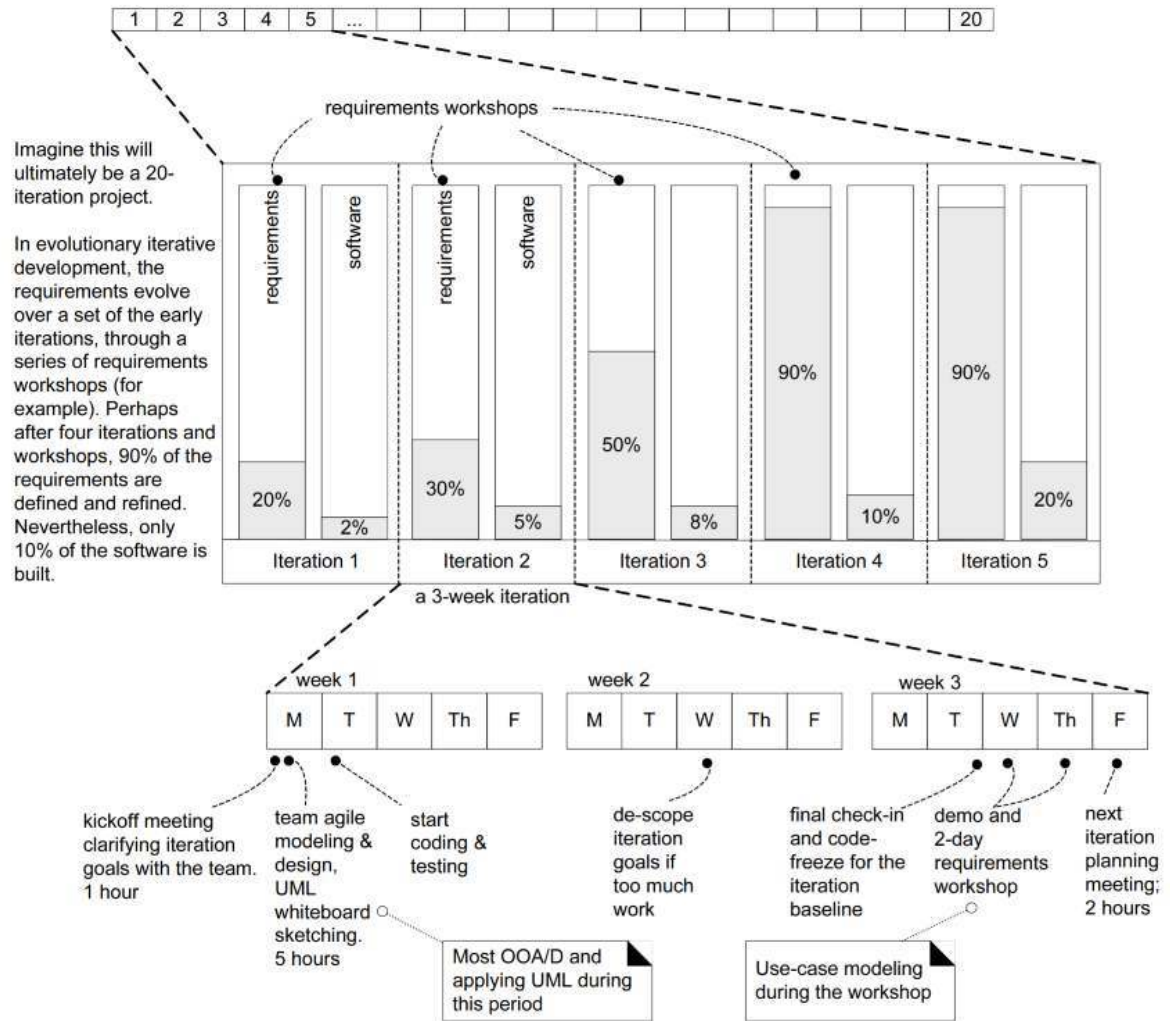
- ▣ Iterations may be considered a series of structured **build-feedback-adapt** cycles
- ▣ Main objectives:
 - ▣ Address changes
 - ▣ Minimize risks

Unified Process (UP)

- The design and requirements instability lowers over time



Iterative and Evolutionary Analysis and Design



Benefits of Unified Process (UP)

- ▣ Less project failure
- ▣ Better productivity
- ▣ Lower defect rates
- ▣ Early rather than late mitigation of high risks
(technical, requirements, objectives, usability, ...)
- ▣ Early visible progress

Benefits of Unified Process (UP)

- Early feedback, user engagement, and adaptation, leads to a refined system that meets the real needs of the stakeholders
- Managed complexity – development team is not overwhelmed by “analysis paralysis” or very long and complex steps
- The learning within an iteration can be methodically used to improve the development process itself, iteration by iteration

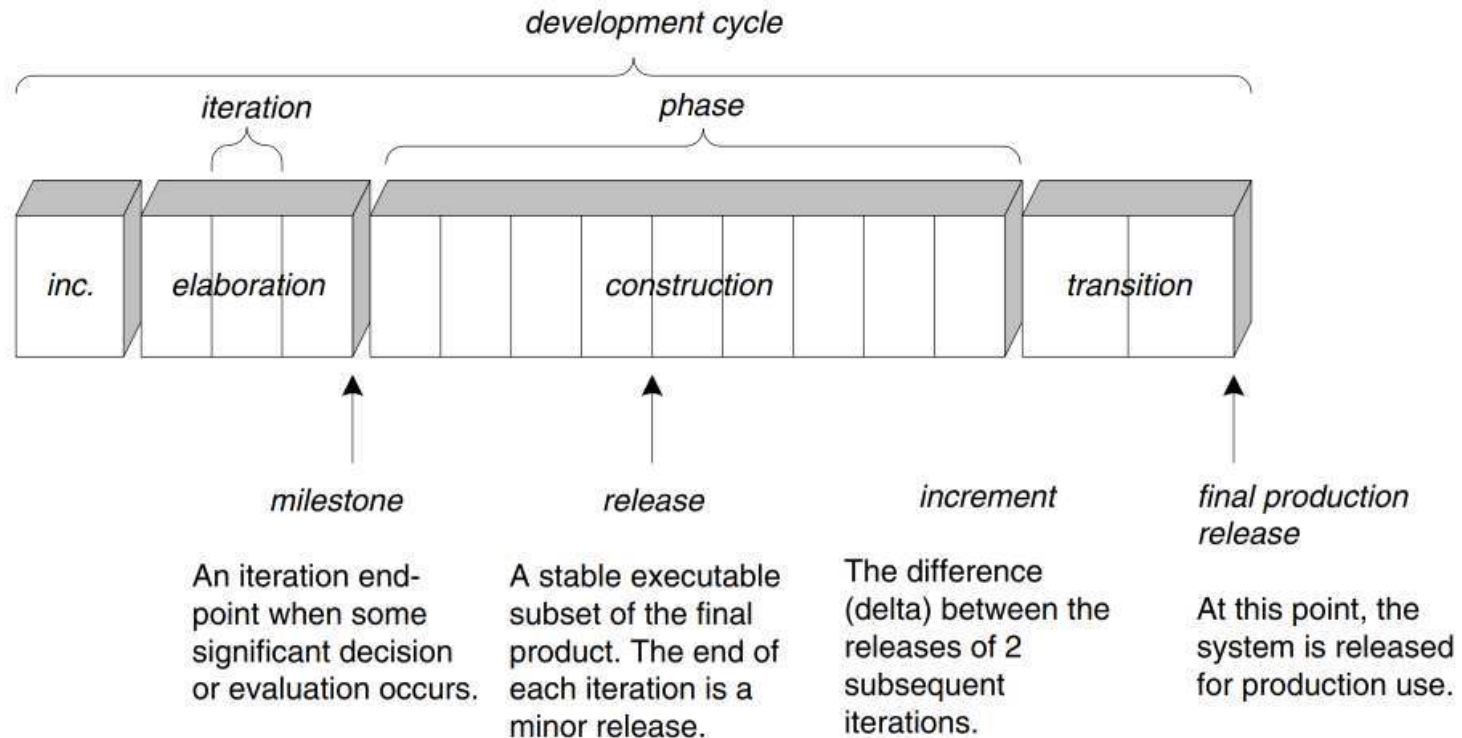
Agile UP

- ▣ Keywords: adaptability and lightness
- ▣ Examples:
 - ▣ Requirements and designs are not completed before implementation; they adaptively emerge through a series of iterations, based on feedback
 - ▣ UML applied with agile modeling practices
 - ▣ No detailed plan for the entire project
 - ▣ ...

UP 4 Phases

- ▣ **Inception:** approximate vision, business case, scope, vague estimates
- ▣ **Elaboration:** refined vision, iterative implementation of the core architecture, resolution of high risks, identification of most requirements and scope, more realistic estimates
- ▣ **Construction:** iterative implementation of the remaining lower risk and easier elements, and preparation for deployment
- ▣ **Transition:** tests, deployment

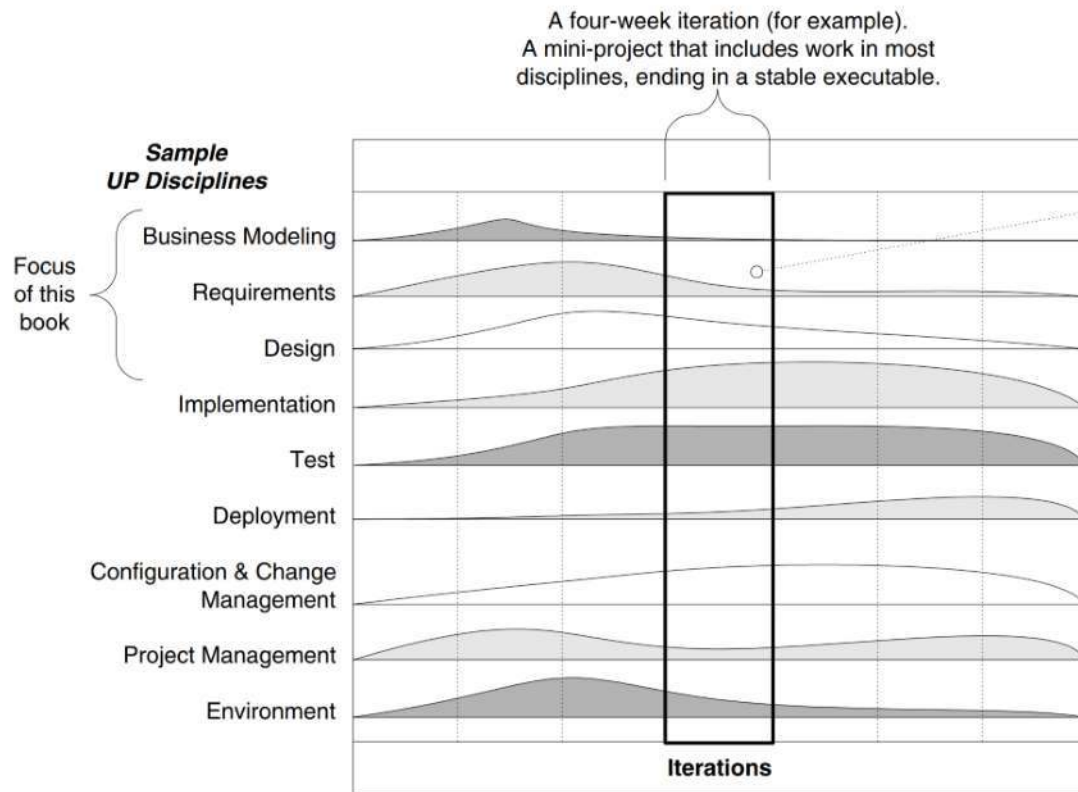
UP Iterations and Phases



UP Disciplines

- ▣ **Disciplines: a set of activities** (and related artifacts) in one subject **area** (e.g., activities within requirements analysis)
- ▣ In UP, an **artifact** is the general term for any **work product**: code, Web graphics, database schema, text documents, diagrams, models, ...

UP Disciplines



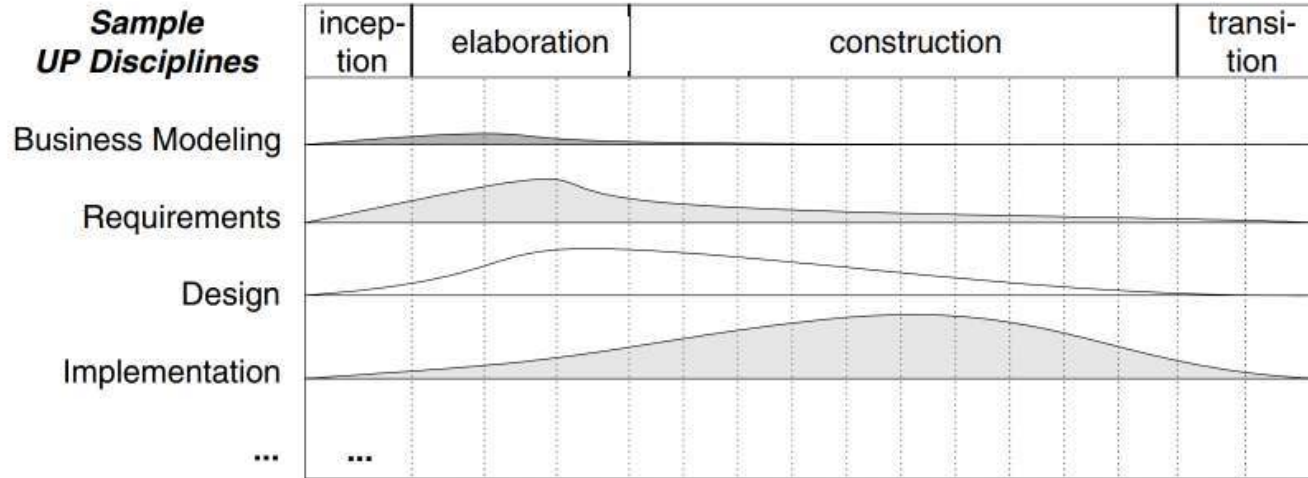
Note that although an iteration includes work in most disciplines, the relative effort and emphasis change over time.

This example is suggestive, not literal.

UP Disciplines and Phases

- ▣ During one iteration work goes on in most or all disciplines
- ▣ The relative effort across these disciplines changes over time
- ▣ Early iterations focus on requirements and design
- ▣ Later iterations, as the requirements and core design stabilize through a process of feedback and adaptation, focus on implementation and testing

UP Disciplines and Phases



The relative effort in disciplines shifts across the phases.

This example is suggestive, not literal.

Part 2: Rational Unified Process

Rational Unified Process (RUP)

- ▣ A detailed refinement of UP
- ▣ The result of each iteration is an executable but incomplete system; it is not ready to deliver into production.
- ▣ The system may not be eligible for production deployment until after many iterations; for example, 10 or 15 iterations
- ▣ UML based analysis and modeling

Part 3: Exercises

Incremental vs Iterative

- ▣ Similarities/Differences?
- ▣ Advantages/Limitations?
- ▣ A possible answer at:
 - ▣ <https://www.plutora.com/blog/iterative-incremental-development-comparison>

Examples of Software Systems – What SEP to Use?

▣ Example 1: **Software Design Project**

- ▣ Teams of 3/4 students
- ▣ Time: 2 months

▣ Example 3: **Web Application for a New University**

- ▣ Teams of 20 software engineers
- ▣ Time: 6 months

▣ Example 2: **Air Traffic Management**

- ▣ Teams of 100 software engineers distributed all over the world
- ▣ Time: 2 years

▣ Example 4: **Lego Store**

- ▣ Teams of 50 software engineers distributed all over the world
- ▣ Time: 1 year

To Do



Your TO DO List for the 2nd Lecture:

- ▣ Read the study material

Reading – For the 2nd Lecture

□ Exam material:

- Ian Sommerville, Software Engineering, 9th or 10th edition – Chapter 2.4
- RUP: RationalUnifiedProcess.pdf

Takeaways?

